EXP 10 : PHOTODIODE RESPONSE AND SENSITIVITY ANALYSIS USING OPTISYSTEM

OBJECTIVES

1. Design and simulate an optical receiver using PIN and APD photodetectors in OptiSystem.

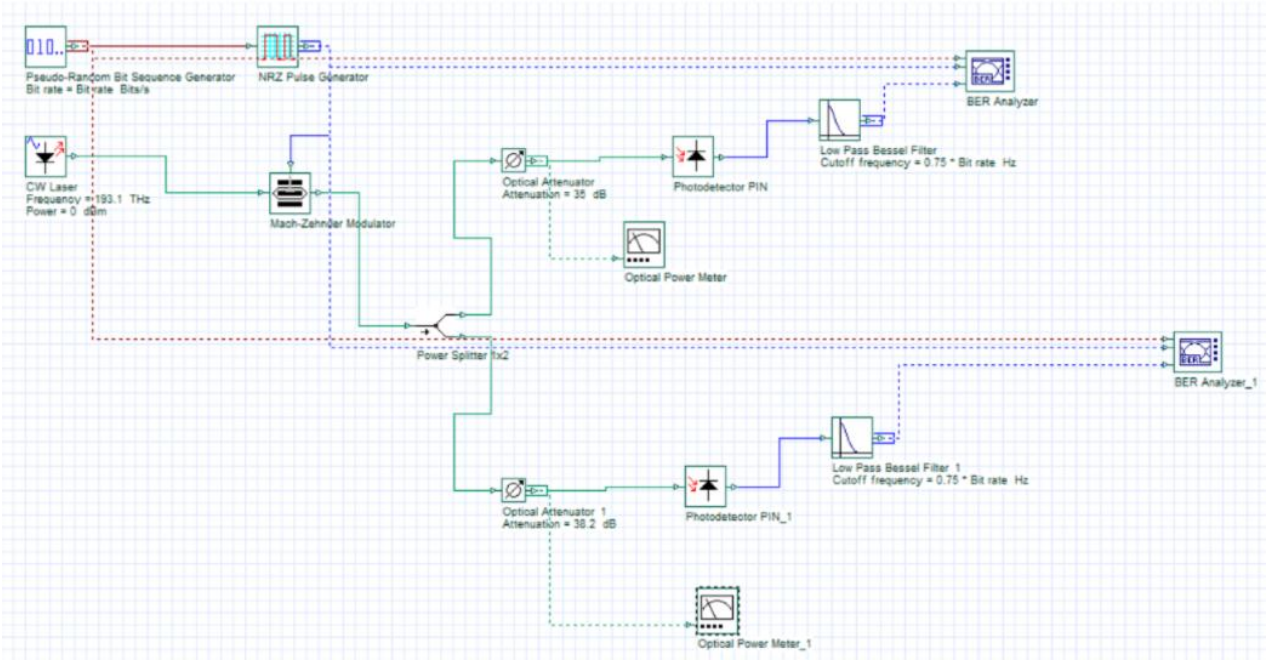
2. Observe the photocurrent generated for a given received optical power and determine the responsivity of the detectors.

3. Determine the receiver sensitivity (minimum received power), bandwidth and the variation of SNR and BER with received power.

4. Compare the PIN and APD receivers and evaluate photodetector performance.

SOFTWARE USED : Optisytem software

CIRCUIT DIAGRAM:



RESULTS:



REFLECTION:

I learned to design and simulate PIN and APD optical receivers in Optisystem and analyze responsivity, sensitivity, bandwidth, SNR and BER. I observed that the APD provides higher photocurrent due to avalanche gain and can operate at lower received optical power that the PIN receiver. modify this with same meaning

RUBRICS:

Objective	**Assessment Criterion**	**Excellent – 4**	**Good – 3**	**Satisfactory – 2**	**Needs Improvement – 1**	**Weightage**
Design the OptiSystem PIN/APD receiver circuit with correct components and parameters.	Circuit Setup and Parameter Definition	Builds the complete circuit (laser, modulator, Fork 1×2, aDenuators, PIN/APD, filters, power meters, BER analyzers) with all parameters correct (193.1 THz, 0 dBm, 35/38.2 dB, 0.75 × bit rate), independently.	Builds the circuit with minor connection or parameter errors and limited assistance.	Builds only part of the circuit or needs considerable assistance; has unit/parameter errors (e.g., dB vs dBm).	Unable to build the circuit or set the parameters.	25
Run the simulation and record received power, Q factor and BER for both receivers.	Simulation Execution and Data Acquisition	Runs the simulation and records received power (W and dBm), eye diagram, Q factor, BER and the aDenuation sweep correctly for PIN and APD.	Records all quantities with minor errors or omissions.	Records some quantities with noticeable errors or inconsistencies.	Unable to run the simulation or record the values.	25

Obtain eye diagrams and BER/SNR plots; determine responsivity, sensitivity and bandwidth.	Graph PloDing and Sensitivity Determination	Plots accurate eye diagrams and BER (semi-log) / SNR curves with labelled axes and a BER = 10^{-9} line; sensitivity within ±1 dB, responsivity and bandwidth within 10% of the reference.	Plots the curves with minor errors; sensitivity, responsivity and bandwidth identified with small deviation.	Plots partial or poorly scaled curves; sensitivity, responsivity or bandwidth approximate.	Unable to plot the curves or determine the parameters.	25
Interpret photocurrent–power relation, SNR/BER variation and PIN vs APD performance.	Analysis and Engineering Interpretation	Correctly explains linear photocurrent, noise-limited regions, SNR/BER trend and the APD gain advantage, and links results to detector selection.	Explains the trends with minor errors and reasonable interpretation.	Performs basic interpretation with noticeable errors or limited justification.	Unable to interpret the results.	25

RESULT:

The experiment demonstrates how a photodetector converts received optical power into photocurrent and influences the overall receiver performance. The following observations were obtained:

- PIN Receiver: At a received optical power of −38.2 dBm, the eye diagram remains open. The photocurrent is proportional to the optical power, with a responsivity of approximately 1 A/W.

- APD Receiver: At −41.4 dBm, which is 3.2 dB lower than the PIN receiver, the eye diagram remains open with a higher signal amplitude due to an avalanche gain of approximately 3, improving responsivity and sensitivity.

- Power Dependence: Below the sensitivity threshold, the SNR decreases and the BER approaches 0.5. Above the threshold, the SNR and Q-factor increase significantly, resulting in error-free transmission.

- Bandwidth: The photodetector bandwidth is significantly higher than the bit rate. Therefore, the receiver bandwidth is primarily determined by the low-pass filter, set to 0.75 times the bit rate.

The APD receiver provides better sensitivity but introduces additional avalanche noise and requires a higher bias voltage. In contrast, the PIN photodiode offers a simpler structure and lower noise at higher optical power. These results are useful for selecting a suitable

photodetector based on the optical power budget and receiver requirements.